

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

18th May 2023

**Subject CP1 – Actuarial Practice
(Paper A)**

Time allowed: 3 Hours 15 Minutes (14.45 - 18.00 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. *Please read the instructions inside the cover page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
2. *Attempt all questions beginning your answer to each question on a separate sheet.*
3. *Mark allocations are shown in brackets.*
4. *Please check if you have received complete Question Paper and no page is missing. If so kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately. You are not allowed to carry the question paper in any form with you.

- Q. 1)** AD is an international conglomerate, that has been in the news recently due to an adverse research report from an international broking firm. The report had raised serious concerns around governance and capital adequacy of the company. ADGreens is one of its subsidiaries which is working in green energy space, which is a capital-intensive business.

Outline why will ADGreens need capital?

[6]

- Q. 2)** What are the 3 principles of Insurance and Pensions? Briefly describe them.

[6]

- Q. 3)** The Government of a developing country which has adopted a New Pension Scheme (NPS) has received a proposal to restore Old Pension Scheme (OPS) for their employee pension benefits against prevailing New Pension Scheme (NPS).

The benefits under both the scheme are:

Old Pension Scheme: 50% of last drawn salary payable as pension until death of pensioner.

New Pension Scheme: Fixed sum deposited by employer and employee into separate fund that accumulates and becomes available on superannuation to purchase annuity.

You are required to:

- i)** Identify & briefly explain the type of pension scheme OPS & NPS represent. (2)
 - ii)** Discuss why Government should continue New Pension Scheme (NPS)? (5)
 - iii)** Comment on implications of the proposal on various stakeholders. (4)
- [11]**

- Q. 4)** An automobile company is contemplating introducing complimentary 4 years from the date of purchase / 100,000 Kms warranty program to its car owners as against 2-year standard warranty offered by other Auto companies. Under this warranty program, car owner shall be entitled to repairing for any of the manufacturing defects related to the paint work and bodywork corrosion.

- i)** Enumerate the practical difficulties in ascertaining implied cost of elongated warranty. (2)
- ii)** What information shall be required to determine provision for unexpired warranty period? (3)

The automobile company has approached ABC General Insurance to insure the risks for its complimentary 4 years warranty program. You are the Pricing Actuary of ABC and your Appointed Actuary has asked you to help design and price the product for discussion.

- iii)** Outline what all aspects you will consider while designing and pricing the product. (5)
- iv)** Your CRO has also asked you to mention the key risks that cannot be priced for. (3)

[13]

- Q. 5)** Post historic loose monetary policy in the world, one of the economically developed nations has faced high levels of inflation. The central bank of the country in its fight against inflation started raising the interest rate from near zero to 475 bps within a span of one year. In view of this, certain banks have begun to fail/face the risk of imminent failure.
- i)** Under given circumstances why would a bank face the risk of failure? (6)
 - ii)** One of the banks is facing liquidity crunch. It has a home loan book that would generate regular cashflows in the form of repayments. How can bank use this loan book to improve its immediate liquidity position and elaborate consequences thereof. (4)
 - iii)** A bank needs capital to expand its business and meet sudden capital needs. Promoters of bank are not keen to dilute their shareholding/infuse further capital/change dividend policy. How can a Bank raise capital under given circumstances and also comment on potential challenges that bank might face. (4)
- [14]**
- Q. 6)** Long term care insurance is a segment that your company is targeting to enter to expand the company's footprint in the country.
- i)** Define Long term care and what needs of a customer does it meet? (4)
 - ii)** You are the Pricing Actuary tasked with doing the initial pricing of the product. You are asked to present the pricing results to the Product Management Council. Discuss the aspects you will consider while deciding on the assumptions and other factors that will have to be taken into consideration while pricing the Long Term Care product? (8)
 - iii)** Longevity risk is one of the key risks in this product. Discuss the scenario(s) when the company should consider it as a systematic or a diversifiable risk? (3)
- [15]**
- Q. 7)** Mr. A is a High Networth Individual (HNI) who wishes to diversify his investment portfolio across various asset classes. Mr. A's portfolio manager has recommended Gold as an asset class. Now Mr. A is inclined to earmark 5% of his corpus in Gold. At present, one can derive exposure to Gold in various ways like Physical Gold, Gold ETF units issued by Mutual funds & Sovereign Gold Bonds issued by Central Government.
- i)** Comment on rationale for investing in gold. (2)
 - ii)** Explain how each of the aforesaid instruments operate. Compare & contrast the characteristics of each of the instruments such that Mr. A can decide the most appropriate instrument. Mr. A, is taxed at maximum marginal tax rate on interest income. (10)
 - iii)** Identify any other instruments that can offer exposure to Gold. (3)
- [15]**
- Q. 8)** A Regulator in country X has asked all the insurance companies to help increase the insurance penetration in the country. To do so, it has asked each of the big insurance companies to pick up certain states of country X and drive various initiatives to help improve the insurance penetration.

- i)** Outline the areas of help that companies would need from the regulator. (3)

You are the Head of Strategy and CEO has asked you to prepare a plan to achieve the objectives.

- ii)** What all aspects will you consider while making the plan? (6)

You have spoken to different departmental heads for their inputs. CRO has suggested that company should have a different product from the existing products for cities in semi urban areas.

- iii)** Why do you think CRO has suggested for a different product? (5)

- iv)** CRO has suggested that product design, sales practices, Under Writing and other aspects of the business may also need a review while expanding into these semi-urban areas. Outline the areas that needs to be considered to ensure risks to the company are mitigated. (6)

[20]
